



Hungarian-Japanese Wordle Dictionary

A Lexicographic Concept

Combining a Wordle-Solving Assistant with a Bilingual Dictionary

MODULE

Dictionary Planning and Production

PROJECT TEAM

Zillan Bunaya Wildan and Partner

Project Overview & Concept

Bridging Gaming and Lexicography for Hungarian & Javanese

CORE CONCEPT

A dual-purpose online platform combining a **Wordle-solving assistant** with a comprehensive **bilingual dictionary**.



Primary Goal

Support Wordle players while facilitating language learning for Hungarian speakers learning Javanese and vice versa.



Target Audience

Language learners, linguists, and word game enthusiasts navigating two distinct linguistic systems.



Key Outcome

A living, bidirectional dictionary where gameplay drives vocabulary discovery and database growth.

M A N U K

 Hungarian



 Javanese

SCOPE

5-Letter Words

Frequency-based selection

Core Functionality & Features

From Wordle Mechanics to Lexicographic Depth



Dual-Database System

Separate, optimized lexical records for Hungarian and Javanese vocabularies operating in parallel.



Wordle Feedback Logic

Inputs constraints (Green/Correct, Yellow/Misplaced, Gray/Absent) to generate valid candidate lists.



Seamless Transition

Solving the puzzle unlocks the full dictionary entry with senses, usage examples, and register details.



Bidirectional Linking

Instant navigation between Hungarian lemmas and their Javanese counterparts (HU ⇄ JV).

USER JOURNEY FLOW

1. Input Constraints



2. Candidate Generation



System filters vocabulary based on constraints



3. Solution Identified



Correct word found (e.g., MANUK)



4. Dictionary Entry



manuk (*n.*)bird
See also: madár (HU)

Development Workflow

Phases 1-2: Preparation, Planning & Data Acquisition

Preparation & Planning

PRE-LEXICOGRAPHY
STAGE

- **Market Analysis:**
Identifying needs of
HU/JV learners &
Wordle players
- **Team Setup:**
Lexicographers,
Computer Scientists,
Corpus Linguists
- **Style Guide:** Defining
Japanese register
treatment
(Ngoko/Kromo)
- **Nomenclature:**
Frequency-based
selection of 5-letter
words

1

Data Acq. & Computerization

DIGITAL INFRASTRUCTURE
SETUP

- **Corpus Compilation:**
Assembling authentic
texts for both languages
- **Social Levels:** Ensuring
Japanese corpus covers
various registers
- **DWS Installation:** Setting
up Dictionary Writing
System with XML support
- **Engine Logic:**
Programming the
Wordle-solving algorithm

2

Development Workflow

Phases 3-4: Analysis & Transfer Stages

The Analysis Stage

MONOLINGUAL PHASE

- **Lexical Analysis:** Using Corpus Query Systems (CQS) like Sketch Engine to identify word senses, collocations, and grammatical patterns.
- **Database Building:** Creating structured records for each lemma (e.g., Javanese *manuk* with specific features) before bilingual transfer.
- **Unbiased Recording:** Establishing comprehensive monolingual datasets for both Hungarian and Javanese independently.

3

The Transfer Stage

BILINGUAL PHASE

- **Equivalence Mapping:** Translators find Target Language (TL) equivalents for Source Language (SL) entries.
- **Handling Lexical Gaps:** Formulating descriptive paraphrases or "explanatory equivalents" where direct matches don't exist.
- **Register Alignment:** Pairing Javanese Kromo/Ngoko distinctions accurately with appropriate Hungarian stylistic variants.

4

Development Workflow

Phases 5-6: Synthesis, Integration & Online Release

Synthesis & Integration

COMPILATION STAGE

- **Entry Finalization:** Extracting data from translated database & adhering to Style Guide.
- **UI/UX Integration:** Merging Wordle feedback loop (colors) with dictionary lookup.
- **Cross-References:** Establishing bidirectional links for immediate HU/JV access.
- **Trigger System:** Full dictionary display triggers upon correct word identification.

5

6

Online Release & Maintenance

CIRCULAR MAINTENANCE

- **Modular Release:** Launching in parts (e.g., 5-letter verbs first) to allow organic growth.
- **Quality Assurance:** Extensive testing of search logic, hyperlinks, and audio.
- **Feedback Loop:** Gathering player input to correct errors & add vocabulary.
- **Living Tool:** Ensuring the dictionary remains dynamic through continuous updates.

Interface Demonstrations

Visualizing the Bilingual Wordle Experience



Select the language of the word you are looking for:

Magyar



Hungarian Language

Jawa



Javanese Language

Please choose a language to begin!

Interface Demonstrations

Visualizing the Bilingual Wordle Experience

Szűrés

Ismert Betűk (Zöld)

M

O

N

O

K

Sárga Betűk (Rossz Hely)

Elvetett Betűk (Szürke)

Keresés

Lehetséges Megoldások

manuk

manuk (madár)

Interface Demonstrations

Visualizing the Bilingual Wordle Experience

Panyaring

Aksara Sing Wis Weruh (Ijo)

M

O

D

O

R

Aksara Kuning (Panggonan Salah)

Aksara Dibuwang (Klawu)

Golek

Solusi Sing Mungkin

madár

madár (manuk)

Interface Demonstrations

Visualizing the Bilingual Wordle Experience

ma·dár

[mɒda:r]

sub.; -ak ; **manuk** (Ngoko), **peksi** (Kromo) : *kéwan mawa swiwi lan wulu sing bisa mabur* : A madarak repülnek. || **manuk-manuké padha mabur.** || *Syn.:* szárnyas || *K-:* -fészek || > szárny

Dictionary Entry Structure

Segment-by-Segment Analysis: Hungarian Lemma 'madár'

SIMULATED USER VIEW

ma·dár [mɒdár] *sub.*; -ak

1. manuk (Ngoko), peksi (Kromo): kéwan
mawa swiwi...

: A madarak repülnek // manuk-
manuké...

Syn.: szárnyas || K-: -fészek > szárny

HUNGARIAN



Targeting Javanese Learners

Schema Breakdown

Database View

SEGMENT	SYMBOL	TERMINUS	FUNCTION / DESCRIPTION
ma·dár	[LZGA]	Lemmazeichengestaltangabe	The written form of the headword.
.	[STrA]	Silbentrennungsangabe	Indicates syllable division.
[mɒdár]	[PhonA]	Phonetische Angabe	Phonetic transcription (IPA).
sub.	[WAA]	Wortartangabe	Part-of-speech indicator (noun).
-ak	[PLbA]	Pluralbildungsangabe	Indicates plural suffix (madarak).
manuk	[ÄzA]	Äquivalentangabe	JV translation equivalent (Ngoko).
(Ngoko)	[StA]	Stilschichtangabe	Register marking (informal/common).
peksi	[ÄzA]	Äquivalentangabe	JV translation equivalent (Kromo).
(Kromo)	[StA]	Stilschichtangabe	Register marking (polite/formal).
kéwan...	[BPA]	Bedeutungsparaphrase	Definition written in Javanese.
A madarak...	[KBeIA]	Kompetenzbeispiel	Sample sentence in source language (HU).
manuk-manuké...	[ÄKBeIA]	Äquiv. Kompetenzbeispiel	Javanese translation of example.

Dictionary Entry Structure

Segment-by-Segment Analysis: Hungarian Lemma 'madár'

SIMULATED USER VIEW

ma·dár [mɒdɒr] *sub.*; -ak

1. manuk (Ngoko), peksi (Kromo): kéwan
mawa swiwi...

: A madarak repülnek // manuk-
manuké...

Syn.: szárnyas || K-: -fészek > szárny

HUNGARIAN



Targeting Javanese Learners

Schema Breakdown

Database View

SEGMENT	SYMBOL	TERMINUS	FUNCTION / DESCRIPTION
sub.	[WAA]	Wortartangabe	Part-of-speech indicator (noun).
-ak	[PlbA]	Pluralbildungsangabe	Indicates plural suffix (madarak).
manuk	[ÄzA]	Äquivalentangabe	JV translation equivalent (Ngoko).
(Ngoko)	[StA]	Stilschichtangabe	Register marking (informal/common).
peksi	[ÄzA]	Äquivalentangabe	JV translation equivalent (Kromo).
(Kromo)	[StA]	Stilschichtangabe	Register marking (polite/formal).
kéwan...	[BPA]	Bedeutungsparaphrase	Definition written in Javanese.
A madarak...	[KBeiA]	Kompetenzbeispiel	Sample sentence in source language (HU).
manuk-manuké...	[ÄKBeiA]	Äquiv. Kompetenzbeispiel	Javanese translation of example.
szárnyas	[SynA]	Synonymenangabe	Semantically related term in Hungarian.
-fészek	[KompA]	Kompositumsangabe	Common compound (madárfészek).
> szárny	[VerwZA]	Verweiszielangabe	Cross-reference target ("wing").

Interface Demonstrations

Visualizing the Bilingual Wordle Experience

ma·nuk

[ma'nuk]

fn; -2 ; **madár** (közb.), **szárnyas** (irod.) : *tollas, szárnyas, repülni képes gerinces állat* : Manuké padha mabur. || **A madarak repülnek.** || *Syn.:* peksi || *K-:* -an || > swiwi

Dictionary Entry Structure

Segment-by-Segment Analysis: Javanese Lemma 'manuk'

SIMULATED USER VIEW

ma·nuk [ma'nʊk] *fn*; -2

1. madár (közb.), szárnyas (írod.): tollas, szárnyas...

: Manuké padha mabur || A madarak repülnek

Syn.: peksi || K-: -an > swiwi

ORIGINAL SCRIPT

ꦩꦤꦸꦏ꧀

"Manuk" in Aksara Jawa

Schema Breakdown

Database View

SEGMENT	SYMBOL	TERMINUS	FUNCTION / DESCRIPTION
ma·nuk	[LZGA]	Lemmazeichengestaltangabe	The written form of the Javanese headword.
.	[STrA]	Silbentrennungsangabe	Indicates Javanese syllable division.
[ma'nʊk]	[PhonA]	Phonetische Angabe	Phonetic transcription (IPA).
fn	[WAA]	Wortartangabe	Part-of-speech indicator (noun).
-2	[PLbA]	Pluralbildungsangabe	Notation for reduplication plural (manuk-manuk).
madár	[ÄzA]	Äquivalentangabe	Standard Hungarian translation equivalent.
(közb.)	[StA]	Stilschichtangabe	Style marking ("common/colloquial").
szárnyas	[ÄzA]	Äquivalentangabe	Formal/literary Hungarian equivalent.
tollas...	[BPA]	Bedeutungsparaphrase	Definition written in Hungarian.
Manuké...	[KBeIA]	Kompetenzbeispiel	Sample sentence in source language (JV).
A madarak...	[ÄKBeIA]	Äquiv. Kompetenzbeispiel	Hungarian translation of example sentence.

Dictionary Entry Structure

Segment-by-Segment Analysis: Javanese Lemma 'manuk'

SIMULATED USER VIEW

ma·nuk [ma'nʊk] *fn*; -2

1. madár (közb.), szárnyas (írod.): tollas, szárnyas...

: Manuké padha mabur || A madarak repülnek

Syn.: peksi || K-: -an > swiwi

ORIGINAL SCRIPT

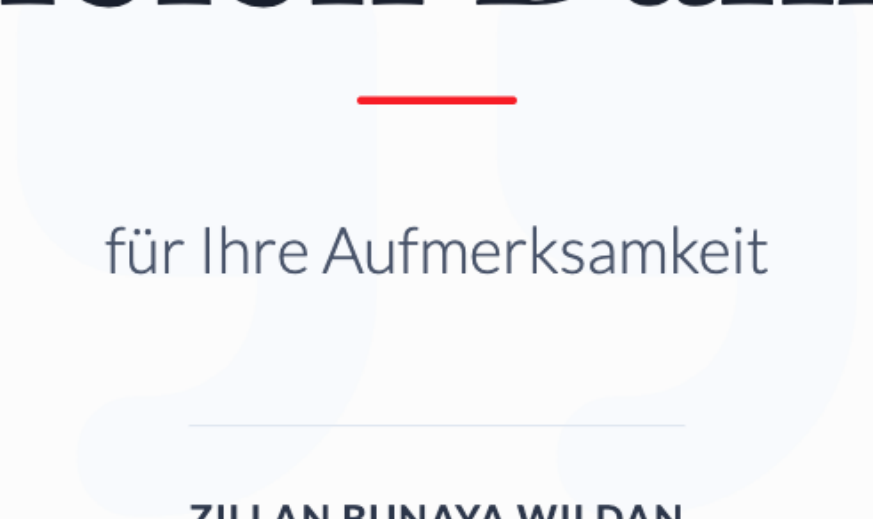
ꦩꦤꦸꦏ꧀

"Manuk" in Aksara Jawa

Schema Breakdown

Database View

SEGMENT	SYMBOL	TERMINUS	FUNCTION / DESCRIPTION
fn	[WAA]	Wortartangabe	Part-of-speech indicator (noun).
-2	[PLbA]	Pluralbildungsangabe	Notation for reduplication plural (manuk-manuk).
madár	[ÄzA]	Äquivalentangabe	Standard Hungarian translation equivalent.
(közb.)	[StA]	Stilschichtangabe	Style marking ("common/colloquial").
szárnyas	[ÄzA]	Äquivalentangabe	Formal/literary Hungarian equivalent.
tollas...	[BPA]	Bedeutungsparaphrase	Definition written in Hungarian.
Manuké...	[KBeiA]	Kompetenzbeispiel	Sample sentence in source language (JV).
A madarak...	[ÄKBeiA]	Äquiv. Kompetenzbeispiel	Hungarian translation of example sentence.
peksi	[SynA]	Synonymenangabe	Javanese Kromo (formal) synonym.
-an	[KompA]	Kompositumsangabe	Javanese compound suffix (e.g., manuk-an).
> swiwi	[VerwZA]	Verweiszielangabe	Cross-reference target ("wing").



Vielen Dank.

für Ihre Aufmerksamkeit

ZILLAN BUNAYA WILDAN

Károli Gáspár Református Egyetem

